Robust Bimanual Handover

CSCI 645 Robot Manipulation, Fall 2026 | Yan Wen

Final project synthesis and CPU review supplement | November 10, 2026

1. Research question and contribution

Can a learned residual improve a geometric controller for contact-rich object handover, and what happens when the object is not exactly where that controller assumes it is? This project implements and evaluates two Franka arms handing over a rigid bar in Isaac Sim using Isaac Lab and PhysX. Each arm uses a two-finger parallel-jaw gripper. There is no dexterous hand, external demonstration dataset or real-robot experiment.

The contribution is an evaluated control pipeline, stronger baseline comparisons, controlled changes to observation and grasp uncertainty, and a reproducible account of where learning helps or fails. A positive PPO result was a hypothesis, not a completion criterion. The main V4 finding is that a calibrated geometric controller configuration handles unknown initial grasp translation well, while the tested learned residuals do not establish an improvement on the primary mixed-ID comparison and can degrade outside the training support.

This supplement is the current concise interpretation. Full methods, all-condition tables and original figures remain in the archived V4 report. Read that report and its deck with the review corrections. All frozen experiments and published releases remain unchanged.

2. One project, four stages

Stage	Question added	What the evidence supports
V1	Does geometric coordination plus residual PPO work?	Geometry was decisive; selection chose step zero in 8/9 runs, limiting claims about trained residuals.
V2	Did learning help after stronger controls and harder cases?	Residual-only learning beat nominal control, but learned residuals on geometry did not beat strong geometric/constant baselines.
V3	Do translation bias, noise and delay make learning useful?	The known giver-grasp reference produced a ceiling: the strongest baseline and selected combined policies reached 100% on the frozen suite.
V4	What if that grasp transform is initially unknown?	Fixed-transform geometry degrades; calibrated geometry is strong; primary learning gains are not established, and boundary fragility remains.

These stages have different suites and assumptions. Their percentages must not be pooled or treated as a single cross-version leaderboard. In V2, residual-only success was 82.6% versus nominal 65.2%, while combined success was 96.1% versus its geometric parent 96.5% and geometry plus a tuned constant 99.2%. These are historical V2 values, not V4 results. Sources: V2 report and V3 report.

3. V4 task, controls and protocol

The giver initially holds an 18 x 3.5 x 3.5 cm rigid bar. The receiver must grasp it, the giver must release, and the receiver must hold independently for 2 s within an 8 s episode. A prohibited terminal collision, drop, release timeout or episode timeout prevents success. Invalid resets count as failures in the all-scheduled denominator. Physics runs at 120 Hz and control at 20 Hz.

The controller combines a nominal trajectory with geometric alignment, a shared holding anchor and contact-gated release. PPO can add a bounded XYZ residual, up to 2 cm per axis. Commanded orientations remain fixed. Policies learn from simulator interaction, using RSL-RL PPO; this is not offline learning from a downloaded dataset.

V4 physically changes the initial in-grasp translation. Training uses x in [-5, +15] mm and z in [-10, +10] mm, plus synthetic translation-measurement corruption for robust profiles. Boundary tests include x = -7 mm and z = +/-15 mm; mixed OOD also increases sensor corruption. The offset is static within an episode: dynamic slip is not tested. Controllers receive the configured sensor/proprioceptive channels; hidden sampled offsets are reserved for initialization and evaluation. This is controlled simulated sensing, not a learned vision system or a complete real-world sensing model.

Controller	Reference / estimator	EMA alpha	Alignment bound
Historical fixed geometry	Known nominal giver transform	1.0	15 mm
EMA geometry	Filtered measured translation	0.2	25 mm
Calibrated geometry	0.5 s measured offset plus giver reference	0.2	25 mm
Combined PPO variants	Same calibrated parent plus residual	0.2	25 mm

The calibration comparison changes multiple settings. Its final-suite effect estimates the complete controller configuration; it does not isolate calibration alone. The strongest constant baseline selected on development data is zero residual and aliases calibrated geometry. It is not a separately executed replication.

Five learned profiles use five training seeds each: clean feed-forward, robust feed-forward, robust four-frame history, robust repeated-current-frame control, and robust residual-only history. Each of 25 runs completes 10,027,008 transitions. The five eligible nonzero checkpoints are ranked on validation only; step zero is a separate control. Selected and final checkpoints are reported separately.

The final suite contains 12 conditions x 200 paired base cases. There are 54 executed entries (129,600 episodes) and 55 logical entries (132,000 including the alias). The primary condition is mixed ID. The two primary contrasts compare selected robust history against robust feed-forward and the strongest constant baseline. The frozen analysis uses 10,000 paired training-seed/base-case bootstrap draws, with 95% and Bonferroni 97.5% intervals. Other conditions and interventions are exploratory.

4. Main results and what they mean

Success percentages below are generated from the published summary, independently checked against raw final records. Learned rows average the five selected checkpoints. Grasp-15 uses x = +15 mm and z = +/-10 mm; the discrete severity grid does not locate a precise continuous x-only failure threshold.

Controller	Grasp-15	Mixed ID	Mixed OOD	x = -7 mm
Fixed geometry	5.5	69.5	73.5	96.0
Calibrated geometry	94.5	98.5	94.5	96.5
EMA geometry	94.0	99.0	90.5	96.0
Robust feed-forward	96.5	99.4	94.5	88.2
Robust history	99.1	100.0	97.0	65.2
Repeated current frame	96.2	99.5	91.3	91.2
Residual-only history	40.3	56.3	53.7	42.3
Nominal	38.5	59.5	59.0	48.0

Primary selected mixed-ID contrast	Difference (pp)	97.5% interval (pp)
History minus feed-forward	+0.6	[+0.0, +2.1]
History minus best constant	+1.5	[+0.0, +3.5]

Both primary intervals include zero at their lower endpoint. These results do not establish superiority, and they do not establish equivalence: no equivalence margin was prespecified. A 100% empirical rate on these cases is not guaranteed success on future trials. Five training seeds also give limited information about optimization variability.

Calibrated minus EMA is -0.5 pp [-1.5, 0.0] on mixed ID and +4.0 pp [0.0, 8.5] on mixed OOD (exploratory 95% intervals). Name the subtraction direction explicitly. The fixed-to-calibrated mixed-ID contrast is +29.0 pp [23.0, 35.5], but includes the change in alignment bound and filtering. The practical result is a better controller package.

Matched development evidence provides a narrower mechanism check. With alpha 0.2 and a 25 mm alignment bound for both controllers, fixed versus calibrated geometry succeeds in 0/24 versus 24/24 grasp-15 cases, 17/24 versus 24/24 mixed-ID cases, and 18/24 versus 22/24 mixed-OOD cases. These cases helped select the baseline and therefore are not independent held-out confirmation of a calibration-only effect.

Selected history reaches 65.2% versus calibrated geometry's 96.5% at x = -7 mm. The paired difference is -31.3 pp [-43.8, -18.7]. This boundary is outside the training support and was absent from checkpoint-selection conditions. Late training is also unstable: final history checkpoints underperform their parent across non-clean conditions. Validation selection mitigates some degradation, but does not guarantee robustness to an unseen boundary.

5. New reset-sensitivity analysis

The independent CPU audit found 380 invalid-reset records across the 54 executed entries, all in the palm-side x = -7 mm condition. Eleven case IDs are invalid in at least one entry and four in every entry. Intersecting validity across all entries leaves 189 common paired cases. Every method is evaluated on exactly that same set.

Selected controller	Common-valid success	Parent success	Difference (pp)
Robust history	68.68%	99.47%	-30.79
Robust feed-forward	91.43%	99.47%	-8.04
Repeated current frame	94.81%	99.47%	-4.66

The common-valid history comparison is -30.79 pp, 95% CI [-43.60, -18.20]. Removing cases that failed initialization for any method therefore does not eliminate the history deficit. This result is a retrospective sensitivity analysis, not a new preregistered test. Intervals are conditional on the retained set, exploratory and unadjusted. Conditioning on reset validity can introduce selection effects; the original all-scheduled result remains primary. No original row was deleted or replaced.

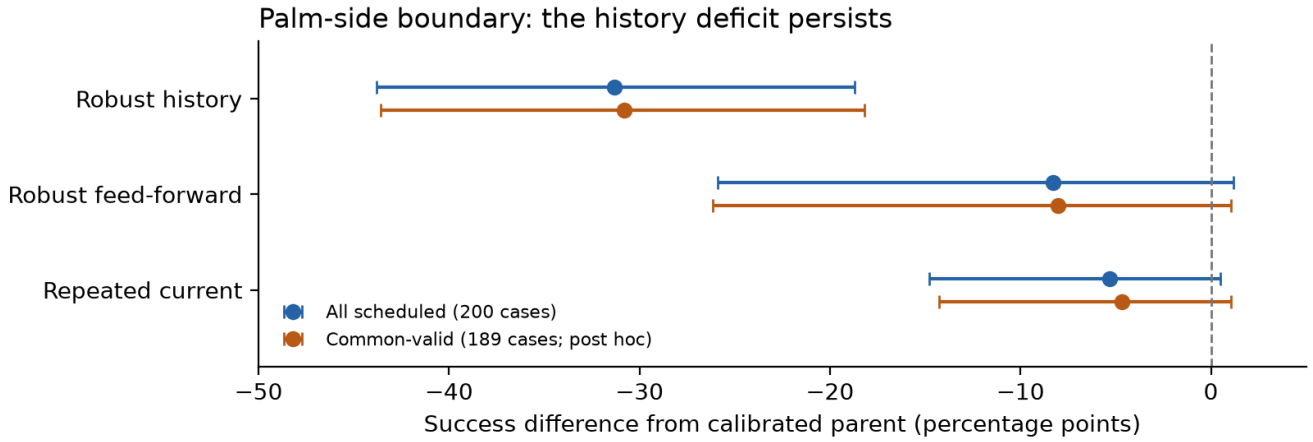


Figure: selected policies minus their calibrated parent, with paired 95% intervals. Blue uses every scheduled case; orange uses the common-valid intersection. These are exploratory boundary comparisons, not the two primary mixed-ID contrasts.

6. Mechanism evidence and limits

On the 60-case mechanism subset, a constant fitted from development residuals has at least as many successes as each selected combined history policy. Reversing history or repeating the current frame changes few outcomes on that subset. Together these observations suggest a substantial constant-offset component and limited demonstrated benefit from temporal order in this setting. They do not prove identical trajectories, that every residual policy is constant, or that memory cannot help a harder task.

The remote repeatability study reports 2,160/2,160 matched terminal outcomes; separate trace runs match 323/324, with one discrepancy. Simulator reproducibility is measured under specified versions and batch configurations, not asserted universally.

7. Reproducibility, delivery and audit boundaries

The v4.0.0 release contains the study checkpoints, raw records, logs, figures, report, slides and videos. The v4.0.1 release repairs a missing verification manifest; use its tooling with v4.0.0 data. Formal experiment source is the immutable experiment-v4 commit `aac4d98274814b1ab121a715db71187e3e1cc0b9`.

The remote public-download verification records 5,007 rehashed archive members, 424 resolved checkpoint index entries, byte-identical regeneration of 35 tables and eight figures, checkpoint reload, small GPU replays and a resume test that preserved original weights. Index entries are not a count of unique weights. The final continuation compute record is 16.18 A100 GPU-hours, including verification.

This CPU review separately downloaded the final-record archive, verified every executed file hash and expected record provenance, reconciled all success/reset summaries, reproduced the two primary bootstrap contrasts, checked 78 frozen Git inputs, and checked 25 checkpoint selections against recorded candidate summaries. It did not independently execute physics, reload weights or re-read raw validation records. See audit evidence, audit code, and remote verification.

The new sensitivity plot and this supplement are generated by `scripts/final_review_build.py`; its measured V4 tables read checked-in JSON outputs. The original reports and releases are preserved. The repository README is the current entry point; historical execution runbooks must not trigger duplicate training.

8. Scope, conclusion and future work

The project tests one rigid object, fixed physical parameters and commanded orientations, static initial translation error and synthetic sensing in simulation. It does not test dynamic slip, arbitrary objects, vision, dexterous hands or transfer to hardware. The calibration window occurs before receiver contact and assumes the grasp offset stays usefully stable. Sensor bias can contaminate that calibration.

Within this scope, strong geometric estimation and coordination explain much of the performance. Residual learning can improve weak nominal control in V2, but the V4 residuals do not establish an advantage over strong baselines on the primary tests and can reduce boundary robustness. That scoped negative result, supported by controls and public evidence, answers the original research question more honestly than an unsupported claim of universal learning benefit.

A future causal calibration study should hold alignment bound and filtering fixed on new cases, with a protocol declared before inspecting their outcomes. Dynamic slip would require a separate physical intervention and acceptance checks. Neither is required to claim completion of the frozen V4 study. The present remaining task is to publish and verify this review supplement, not to retrain the complete matrix.

9. Related work and assistance disclosure

Residual learning combines an existing controller with a learned correction, as in Johannink et al. and Silver et al. PPO supplies the policy optimization method. Frau-Alfaro et al. study learned dexterous object handover; this project deliberately uses parallel-jaw grippers and focuses on controlled geometric and sensing assumptions. These references motivate the approach, rather than implying a reproduction of their tasks or results. Isaac Lab and RSL-RL provide simulation/training infrastructure.

- Johannink et al. (2018 preprint), Residual Reinforcement Learning for Robot Control.
- Silver et al. (2018 preprint), Residual Policy Learning.
- Schulman et al. (2017), Proximal Policy Optimization Algorithms.
- Frau-Alfaro et al. (2025), Learning Dexterous Object Handover.
- Mittal et al. (2025), Isaac Lab: A GPU-Accelerated Simulation Framework for Multi-Modal Robot Learning.
- Schwarke et al. (2025), RSL-RL: A Learning Library for Robotics Research.

Primary arXiv records were checked on November 10, 2026 for title and identity. This supplement uses preprint metadata; it does not independently certify every publisher page range or DOI appearing in older documents.

The archived study discloses AI-assisted implementation, experimentation, analysis and writing through a remote coding agent. Local workflow maintenance and this CPU audit/synthesis were also produced with Codex assistance under the owner's direction. The measured results come from recorded simulator runs, not generated illustrations or fabricated examples. The owner should understand and accurately disclose that assistance when presenting the project.

Reading and presentation package

- Current README: project status, artifact links and execution entry point.
- Full V4 report: detailed methods and all-condition results.
- Review and corrections: interpretation changes and exact audit scope.
- Defense notes: five-minute narrative, questions and demo order.
- Final delivery assignment: remaining publication and fresh-download checks.

The concise supplement and full archived report should travel together. The supplement adds interpretation and sensitivity evidence; it does not silently replace the study's original analysis or hide unfavorable results.